Trainbow HUB — Product Requirements Document (v1.1, Refined)

Oct 10, 2026 · @Micha

Trainbow HUB is a mobile-first learning platform where Nigerian students get AI-guided Mathematics help and book verified tutors in one account. This version tightens v1.0 so a team can estimate, build and test it: the MVP launches with Mathematics on a platform built to add any subject, with one booking path, every requirement has an ID and priority, and open decisions are listed instead of buried.

1. Vision, summary and what changed

Vision. Build a reliable, accessible, AI-assisted education platform that connects students, parents, tutors and learning resources. Tagline: Learn • Connect • Grow.

Primary objective. A learner can register, learn, get personalised help, find a tutor, pay for and attend a lesson, and see progress from one account.

Four service areas, one system. Academy (courses and revision), AI (guided learning assistant), Tutors (directory, booking, virtual lessons, reviews) and Music (theory and instrument tuition). They share one account, one calendar, one payment system and one progress record. At MVP only Academy (Mathematics), AI (Mathematics) and Tutors ship; Music reuses the same foundations later.

Launch approach. Lean, Nigeria-first MVP: Mathematics, a curated tutor marketplace, NGN payments. International readiness is an architecture constraint (currency, time zone, locale fields), not an MVP feature.

Changes from v1.0

Area	Change in v1.1	Why
Scope	MVP defined explicitly with in/out lists	v1.0 listed all four divisions as modules with no clear cut line
Requirements	IDs (e.g. BK-04), priority (M/S/C) and testable wording	Enables estimates, backlog and traceability to tests

Area	Change in v1.1	Why
Guardians	Minimal guardian consent and linking moved into MVP	Minors cannot be served safely without it; v1.0 left the portal as "should have"
Payouts	Manual tutor payouts at MVP, automation later	Marketplace split payouts depend on provider support and compliance
Booking	Separate booking status from payment status, with a defined state table	v1.0 mixed them in one lifecycle
Metrics	Each metric has a measurement method and review owner	v1.0 targets were not measurable as written
Privacy	Anchored to the Nigeria Data Protection Act 2023 and a named compliance owner	v1.0 only said "review applicable law"
Open questions	New decision log with owner and needed-by date	Prevents silent assumptions during build
LMS	Build a light native learning module for MVP; Moodle only if discovery justifies it	Avoids integration cost and a second user system for one subject

2. Goals, scope and success metrics

The MVP succeeds if a pilot cohort completes the full journey (register, learn, book, pay, attend) reliably and safely. Targets below are pilot hypotheses to validate, not promises.

Business goals

1. Extend the existing physical tutoring business into digital delivery.
2. Earn revenue from tutoring commission, subscriptions and courses.
3. Cut admin work through automated scheduling, reminders and reporting.
4. Build a trusted, verified tutor network.
5. Create a brand and architecture that can later serve other markets.

MVP scope

In scope (MVP)	Out of scope (later)
Public site, course and tutor catalogue	Native iOS and Android apps
Registration, roles, guardian consent for minors	Music courses and instrument bookings
Multi-subject catalogue (admins add subjects without code); Mathematics launches first with lessons, quizzes, progress tracking	AI answers for non-Mathematics subjects until each passes evaluation
AI Mathematics assistant (text only) with quotas	Voice or image-based AI input
Tutor profiles, vetting, availability, booking	Automated split payouts to tutors
NGN payments, refunds, reconciliation, tutor ledger	Multi-currency and international payments
Third-party video classroom link	Lesson recording
Email and SMS notifications, admin and support tools	School administration features, multilingual UI

Success metrics

Each metric has a source and a reviewer. The product owner reviews them weekly during the pilot. Baselines are set in the first two pilot weeks, since no digital data exists yet.

Metric	Target	How it is measured
Signup success	⩾ 95% of valid attempts	Funnel events: form started vs account verified
Booking completion	⩾ 90% of started flows end without a system error	Server error rate per booking attempt
Booking accuracy	⩾ 99% of confirmed bookings match tutor calendar and payment	Weekly automated audit query

Metric	Target	How it is measured
Payment reconciliation	100% of confirmed payments matched to an order	Daily reconciliation job vs provider report
Page performance	LCP ⩽ 2.5 s at 75th percentile on mobile	Real-user monitoring
Availability	⩾ 99.5% monthly, excluding agreed maintenance	External uptime monitor
AI answer quality	⩾ 80% positive ratings and ⩾ 95% correct on a reviewed Mathematics sample	User ratings plus tutor-scored rubric sample
Learning engagement	Weekly active learners; share booking a second lesson within 30 days	Product analytics
Learning outcome	Quiz score change over 8 weeks, tracked separately from satisfaction	Assessment attempt data
Safeguarding	100% of high-severity reports triaged within 24 hours	Incident queue timestamps

Assumptions and risks

Item	Type	Mitigation
Existing Trainbow tutors and students seed the pilot supply and demand	Assumption	Confirm cohort size and tutor count before build
Payment provider supports the planned checkout, refunds and webhooks	Assumption	Confirm in discovery; sandbox test before commitment
Users book or pay tutors off-platform after first contact	Risk	Hide direct contact details, add value through reminders, progress records and reviews

Item	Type	Mitigation
AI gives wrong Mathematics answers	Risk	Answer verification, grounded content, rubric review, tutor escalation
AI and video costs exceed unit economics	Risk	Per-user quotas, cost alerts, plan limits
Unreliable mobile data and power	Risk	Lightweight pages, resumable quizzes, clear retry states, SMS reminders
Safeguarding or data-protection failure involving minors	Risk	Consent flow, tutor vetting, incident process, legal review before launch

3. Users, roles and permissions

Seven roles exist, and every permission is enforced on the server; hiding a button is not security. A user can hold several roles (a parent who also tutors), but each action is checked against the role being used.

Role	Can do	Cannot do
Guest	Browse courses, tutor profiles (limited), pricing	See private data, book, use AI
Student	Learn, take quizzes, use AI, book, view own progress and payments	See other learners' records
Parent/guardian	Manage linked children, consent, pay, view reports and attendance	Access unlinked children; read raw AI chats unless the child's settings allow it
Tutor	Manage own profile, availability, bookings, assignments, earnings view	Contact learners outside platform channels; see data of non-booked learners
Content manager	Create and publish courses, lessons, quizzes	See payments or private user data

Role	Can do	Cannot do
Support agent	View tickets and the booking and payment status needed to resolve them	See card data, full AI conversations or impersonate users
Administrator	Manage users, tutors, content, refunds, settings, reports	Act without MFA; edit financial ledger entries

Rules for minors

- **US-01 (M).** Under-18 accounts are created by or linked to a verified guardian before booking, payment or tutor contact is allowed.
- **US-02 (M).** The platform records who gave consent, when, for what purpose, and which version of the notice.
- **US-03 (M).** Learners under 18 cannot message tutors privately; communication runs through monitored booking channels.
- **US-04 (M).** Guardians can review, pause and revoke a child's access, and every change is audit-logged.
- **US-05 (M).** The age threshold and consent method are set by counsel after review of the Nigeria Data Protection Act 2023 and related guidance (see open questions).

4. MVP functional requirements

Priority key: M = must have for MVP, S = should have soon after, C = could have. Each row is testable and maps to a test case.

A. Public website

ID	Requirement	Pri	Acceptance criteria
PW-01	Home, about, contact, FAQ, pricing, course catalogue	M	Visitor finds courses and tutors without registering
PW-02	Searchable tutor directory showing only approved tutors	M	Unapproved or suspended tutors never appear
PW-03	Price, cancellation and refund terms shown before payment	M	Terms visible on tutor page and checkout

ID	Requirement	Pri	Acceptance criteria
PW-04	Terms, privacy notice, safeguarding and refund policies published	M	Linked in footer and at signup
PW-05	Responsive, keyboard-usable, SEO-indexable public pages	M	Passes mobile and keyboard checks; sitemap present
PW-06	No private learner data or tutor contact details published	M	Verified by review and automated checks

B. Identity and access

ID	Requirement	Pri	Acceptance criteria
ID-01	Email or phone signup with OTP verification	M	OTP expires in 10 minutes or less; rate-limited per user and IP
ID-02	Login, logout, password reset, session expiry	M	Passwords hashed; reset links single-use
ID-03	Role-based dashboards and server-side authorisation	M	Changing a URL or API ID returns 403, not another user's data
ID-04	Guardian linking with verification	M	Duplicate or conflicting links are rejected
ID-05	Account suspension, deletion request, data export	M	Suspended users cannot book or pay
ID-06	MFA for admin and support accounts	M	Privileged login impossible without second factor
ID-07	Audit log for sensitive admin actions	M	Who, what, when, before and after values recorded; not editable

C. Learning (Mathematics)

ID	Requirement	Pri	Acceptance criteria
LN-01	Course, lesson and quiz delivery for defined levels	M	Student can enrol, complete and resume a lesson

ID	Requirement	Pri	Acceptance criteria
LN-02	Quiz scoring with stored attempts and feedback	M	Score saved before confirmation is shown
LN-03	Progress persists across devices	M	Phone to laptop shows the same state
LN-04	Failed requests show a clear error and retry; in-progress quiz answers are not lost	M	Simulated network drop preserves answers
LN-05	Revision recommendations from quiz results	S	Suggested topics match weak areas
LN-06	Content manager authoring and publishing workflow	M	Draft, review, publish states with version history

D. Trainbow HUB AI (Mathematics)

The AI is a tutor-style assistant, not a general chatbot. It uses an established model API called from the server, never from the browser. No proprietary model is trained in the MVP.

ID	Requirement	Pri	Acceptance criteria
AI-01	Text Q&A with step-by-step Mathematics explanations	M	Response is readable, structured and labelled as AI-generated
AI-02	Hint-first mode, full solution on request	M	Student can choose hint level before the answer
AI-03	Practice questions with independently verified answers	M	Generated answers checked by a calculator or solver step before display
AI-04	Grounding in approved curriculum content where available	S	Answers cite internal lesson or topic, never invented references
AI-05	Per-user daily quota and abuse controls	M	Over-quota requests blocked with clear message

ID	Requirement	Pri	Acceptance criteria
AI-06	Retry-safe requests	M	Duplicate submissions do not double-count usage or create duplicate records
AI-07	Provider outage handling	M	Clear error, manual retry, no usage deducted for failed calls
AI-08	Input and output moderation, age-appropriate responses	M	Test set of unsafe prompts is blocked or redirected
AI-09	Feedback controls: helpful, not helpful, report error	M	Ratings stored against the turn
AI-10	Escalate to a human tutor when unresolved or low confidence	M	One tap creates a support request or booking prompt
AI-11	Conversation history with retention and delete controls	M	Deletion removes content within the stated retention window
AI-12	AI cannot read other students' data, payments or unrelated chats	M	Prompt context built only from the current learner's allowed data
AI-13	Cost, latency and error monitoring with alerts	M	Daily cost alert threshold configured
AI-14	Pre-release accuracy evaluation against a Mathematics rubric	M	Agreed pass mark met on a reviewed sample before public access

E. Tutor marketplace

ID	Requirement	Pri	Acceptance criteria
TM-01	Tutor onboarding with identity and credential checks	M	Profile stays hidden until an admin approves it
TM-02	Profile: photo, qualifications, subjects, levels, language, format, rate, cancellation terms	M	All fields validated; rate in NGN minor units
TM-03	Filters: subject, level, price, format, language, availability	M	Results match every selected filter

ID	Requirement	Pri	Acceptance criteria
TM-04	Reviews only from learners with a completed paid booking	S	Public submission impossible; one review per booking
TM-05	Tutor availability management with recurring and one-off slots	M	One calendar per tutor, used by all services
TM-06	Tutor conduct policy and suspension workflow	M	Suspension cancels future public bookings with notice

F. Booking

ID	Requirement	Pri	Acceptance criteria
BK-01	Temporary slot hold during checkout (default 10 minutes, configurable)	M	Expired holds release the slot automatically
BK-02	Concurrency-safe confirmation	M	Parallel test: only one of two simultaneous requests wins
BK-03	Booking confirmed only after server-verified payment	M	Browser redirect alone never confirms
BK-04	Duplicate payment events cannot create duplicate bookings	M	Replayed webhook leaves one booking
BK-05	Times stored in UTC, shown in the user's time zone (default Africa/Lagos)	M	Display correct across zones
BK-06	Reschedule re-checks availability	M	Conflicts rejected
BK-07	Cancellation and refund outcome follows the tutor's stated policy and is visible to both sides	M	Outcome recorded and shown
BK-08	Reminders by email, SMS or in-app, per user preference	M	Sent at configured intervals before the session
BK-09	Booking status and payment status stored separately	M	See state table in section 7

G. Online classroom

ID	Requirement	Pri	Acceptance criteria
CL-01	Unique, time-limited session link via an established video provider	M	Only booked participants and the guardian can join
CL-02	Link access revoked on cancellation or expiry	M	Old link fails after status change
CL-03	Attendance recorded against the booking	M	Join and leave times saved
CL-04	Report-conduct control inside the session flow	M	Report reaches the safeguarding queue
CL-05	Fallback contact path if the video link fails	M	Support workflow documented and tested
CL-06	Recording off by default	M	Enabled only with lawful basis, consent and retention rule

H. Payments and tutor earnings

ID	Requirement	Pri	Acceptance criteria
PY-01	Local NGN payments through an approved provider; no card data stored	M	Provider-hosted payment fields
PY-02	Server verifies each transaction by provider API and signed webhook	M	Invalid signature rejected; success page alone insufficient
PY-03	Idempotency keys on payment and booking operations	M	Retried request returns the original result
PY-04	Immutable financial event ledger	M	Corrections made by new entries, never edits
PY-05	Commission calculated from recorded events	M	Rate configured per tutor or globally; test cases pass

ID	Requirement	Pri	Acceptance criteria
PY-06	Refund requested, approved, processed and reconciled as separate steps	M	A request is never shown as a completed refund
PY-07	Daily reconciliation against provider settlement reports	M	Unmatched items flagged for review
PY-08	Manual tutor payouts from the ledger at MVP	M	Payout record references ledger entries
PY-09	Handling of failed payments, partial refunds, chargebacks and disputes	M	Each outcome defined for booking status and tutor earnings
PY-10	AI subscription plans with usage limits	S	Entitlement changes only after verified payment

K. Administration and support

ID	Requirement	Pri	Acceptance criteria
AD-01	User, tutor and content management by role	M	Least-privilege permissions tested
AD-02	Tutor approval queue with documents viewed securely	M	Credential files in private storage with signed links
AD-03	Booking, payment, refund and dispute monitoring	M	Filter by status and date
AD-04	Safeguarding and complaint queues with severity	M	Severity timestamps and assigned owner
AD-05	AI usage and cost report	M	Daily cost per user and total
AD-06	Operational alerts (payment failures, booking errors, downtime)	M	Alert reaches on-call contact

L. Multi-subject support

Mathematics is the first subject, not a special case. Any subject is added through configuration and content, not new code, and the AI is switched on per subject only after it passes evaluation.

ID	Requirement	Pri	Acceptance criteria
SB-01	Subject registry with name, levels, delivery formats and status (draft, active, retired); admins manage it without a deployment	M	A new subject appears in the public catalogue only when Active
SB-02	Courses, lessons, quizzes and resources belong to a subject and level; content managers can be limited to assigned subjects	M	A content manager cannot edit outside their subjects
SB-03	Tutors are approved per subject and level, with credential evidence for each	M	A tutor is listed and bookable only for subjects they are approved for
SB-04	Search, filters, dashboards, calendar and booking work for every active subject	M	Subject filter returns only matching courses and tutors; one calendar shows all subjects
SB-05	AI access is enabled per subject; subjects without AI show courses and tutors instead	M	With the flag off there is no AI entry point and the page explains the alternatives
SB-06	A subject's AI is enabled only after a subject-specific evaluation: rubric, approved grounding content, unsafe-content tests and recorded sign-off	M	The enable switch is blocked until sign-off is recorded
SB-07	Per-subject AI configuration (prompt, guardrails, answer-verification method), versioned	S	Changes are logged and can be rolled back
SB-08	Per-subject reporting: enrolments, bookings, AI usage and cost, ratings	M	Admin can filter every report by subject
SB-09	Optional per-subject pricing and commission overrides	S	Override applies only to that subject and is ledger-audited

Subject launch checklist. A subject goes live when all of these are done:

1. Curriculum and level map agreed.
2. Content published and reviewed.
3. Enough approved tutors, with the minimum number set by the business.

4. AI evaluated and signed off, if AI is enabled for it.
5. Pricing, cancellation terms and commission set.
6. Support staff briefed on the subject.

Open decision. Which subjects and curricula follow Mathematics, and in what order? This sets content, tutor recruitment and AI evaluation work for release 2 (owner: business owner).

5. Post-MVP modules

Modules I and J keep their v1.0 letters. Music and the full parent portal reuse MVP foundations, so each ships only after the MVP passes its release gates and the pilot shows demand.

I. Trainbow HUB Music (release 2)

ID	Requirement	Pri	Acceptance criteria
MU-01	Music catalogue and tutor profiles for instruments, voice and theory, on the same account system	S	No second login or booking system
MU-02	Online and in-person sessions in the shared booking calendar	S	Appears beside Mathematics lessons on one dashboard
MU-03	Practice assignments and tutor progress notes	S	Notes visible to the learner and guardian
MU-04	Licensed materials only	S	Each resource has a licence record
MU-05	Adding new music courses needs no code change	S	Content manager publishes a course end to end

J. Parent and guardian portal (full version, release 1.1)

Basic guardian linking, consent and payment are in the MVP (US-01 to US-04, ID-04). The full portal adds:

ID	Requirement	Pri	Acceptance criteria
PP-01	Child progress and assessment summaries	S	Guardian sees only linked children
PP-02	Attendance, upcoming classes, booking and payment history	S	Matches booking records
PP-03	Notification preferences with no sensitive content in previews	S	Lock-screen previews show generic text
PP-04	Review, revoke and audit guardian access	S	Revocation takes effect immediately
PP-05	Students cannot grant guardian rights without verification	S	Unverified grant attempt fails

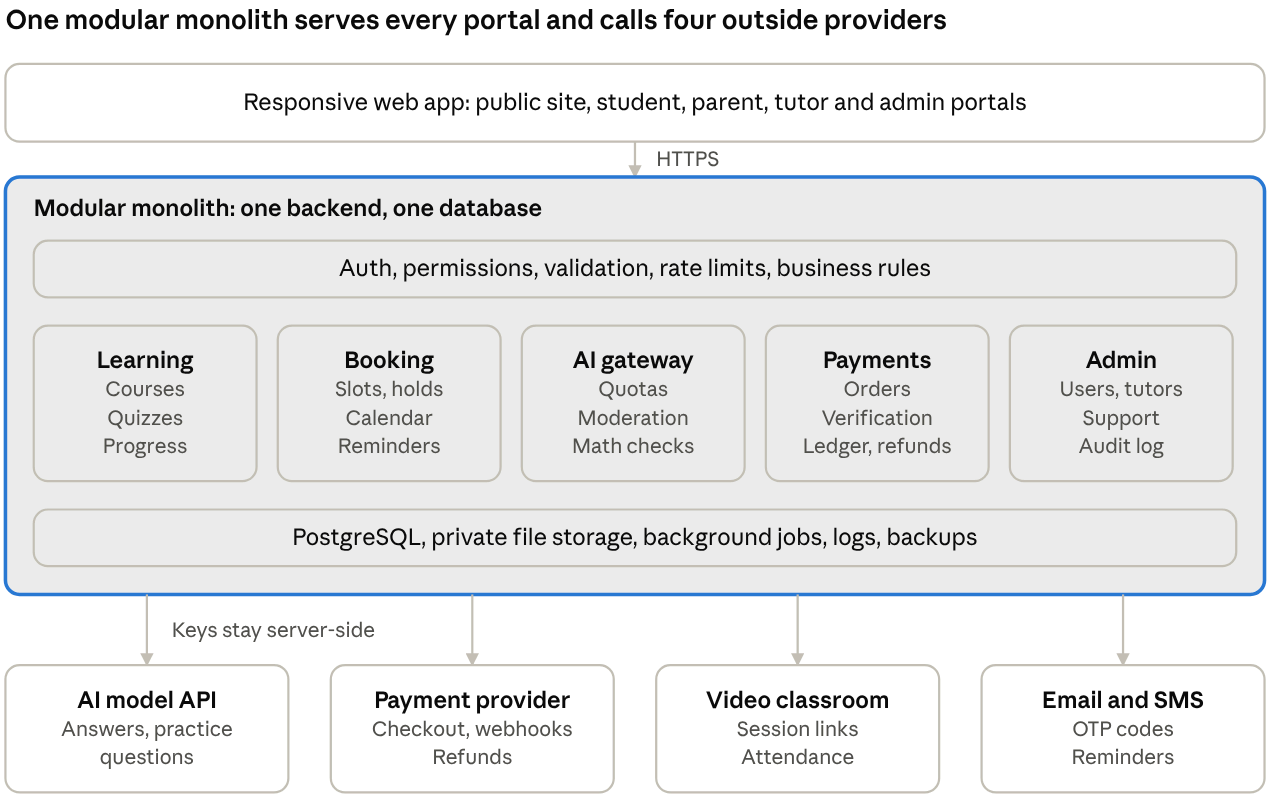
Later

- School administration features and a multilingual interface. (Additional subjects are not deferred: see module L.)
- Progressive web app, then native mobile apps.
- Automated tutor payouts, international payments and specialised AI models, only if usage and cost justify them.

6. Non-functional requirements, architecture and data

Build one modular monolith with a single PostgreSQL database. Separate services are added only when load or team size demands it, because every extra service adds failure points and cost.

Architecture at a glance



architecture · 1 backend, 4 outside providers

The browser never calls an outside provider directly. Every call runs through a server module that holds the keys and checks permissions.

Non-functional requirements

Area	Requirement	Target or test
Performance	Optimised for low-end phones and slow networks; compressed images; no heavy client bundles on public pages	LCP ⩽ 2.5 s (p75) on mobile; page weight budget set in discovery
Availability	Managed hosting, health checks, alerts	99.5% monthly uptime
Security	HTTPS, least privilege, rate limits, dependency scanning, secrets in a vault	Pre-launch penetration test; no open critical or high findings

Area	Requirement	Target or test
Reliability	Transactions, retries with backoff, idempotency, dead-letter handling for jobs	Failure-injection tests for payment, AI and video providers
Accessibility	Aim for WCAG 2.2 AA	Keyboard, screen-reader and contrast audit
Privacy	Collect only what is needed; retention and deletion rules per data type	Data inventory approved by compliance owner
Observability	Central logs, metrics, error tracking with personal data scrubbed	Alert on payment, booking, AI cost and uptime
Recovery	Automated daily backups, point-in-time recovery, written runbook	Restore test passes before launch; RPO and RTO set in discovery
Maintainability	Version control, CI tests, documented APIs, repeatable deploys	Green pipeline required to release
Internationalisation	Currency, locale and time zone stored as data, not hard-coded	NGN and Africa/Lagos defaults; second currency can be added by configuration
Low-bandwidth tolerance	Quiz and form state saved locally and synced; SMS fallback for reminders	Works through a simulated 3G drop

Recommended stack (to confirm in discovery)

- **Frontend and API:** Next.js with TypeScript, or a separate structured API if the team prefers.
- **Database:** PostgreSQL with database constraints for booking overlap and unique payment references.
- **Auth:** managed identity provider or mature framework, with OTP support for Nigerian phone numbers.
- **Files:** private object storage with signed, time-limited links.

- **Learning content:** a light native module for MVP; adopt Moodle only if discovery shows a clear need.
- **AI:** a major model API behind a server-side gateway that handles prompts, quotas, moderation and logging.
- **Payments:** a Nigerian provider with verified webhooks and refunds (for example Paystack or Flutterwave), subject to the discovery checks in PY-02 and PY-08.
- **Video:** an established provider with join links and attendance data.
- **Environments:** development, staging (sandbox payments, synthetic data) and production, each with separate credentials.

Core data model

Entity group	Entities	Key rules
Identity	User, Role, Permission, StudentProfile, GuardianLink, TutorProfile	One user, many roles; guardian link has verification status and revocation date
Learning	Subject, Level, TutorSubject, Course, Lesson, Enrollment, Assessment, AssessmentAttempt	Attempts are append-only; each Subject has a status and an AI-enabled flag
AI	AIConversation, AITurn	Owner-only access; retention date on every turn; cost and token metadata stored
Scheduling	AvailabilitySlot, SlotHold, Booking, ClassroomSession	Exclusion constraint prevents overlapping confirmed bookings per tutor
Money	Order, PaymentTransaction, Refund, TutorLedgerEntry, Subscription	Amounts as integer minor units with explicit currency; ledger entries are immutable
Service	Notification, Review, SupportTicket, AuditEvent	Review tied to a completed booking; audit events are append-only

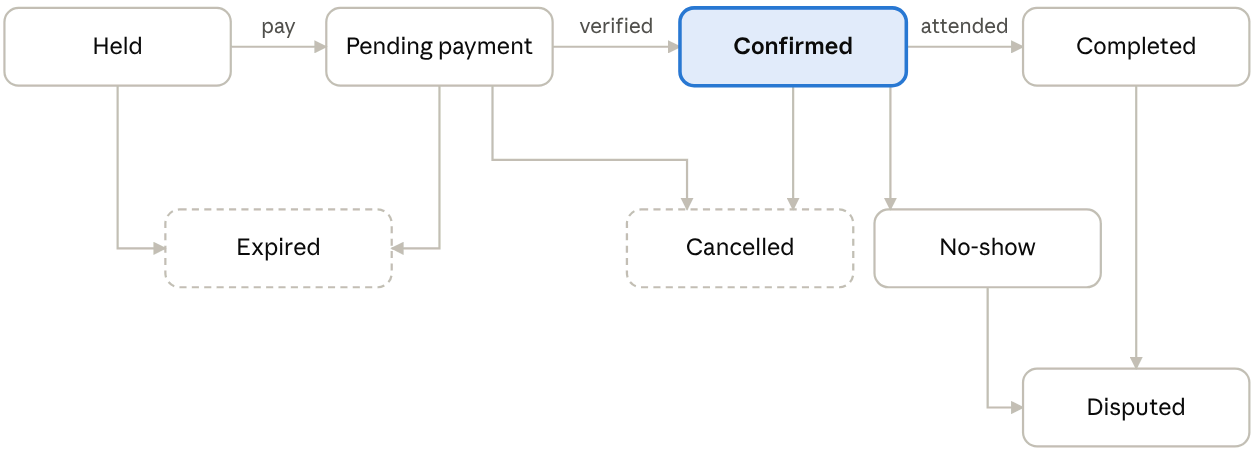
Integrity rules. Every record has a stable unique ID. Financial records reference an order, booking or subscription. Provider transaction references are unique. Tutor earnings are computed from ledger events, never from editable figures. All timestamps are UTC.

7. Key workflows and state rules

Booking and payment are tracked as two separate statuses. A booking is confirmed only when its payment status is Verified, and a refund request never changes the payment to Refunded by itself.

Booking states

Only verified payment moves a held slot to Confirmed



Dashed border = end state. Highlighted = payment verified.

Disputed resolves to Completed or Cancelled

booking states · 8 states

A hold expires if checkout stalls, and a slot becomes Confirmed only after server-side payment verification. Cancellations and no-shows follow the tutor's stated policy.

Status	Meaning	Allowed next states
Held	Slot reserved during checkout	Pending payment, Expired
Pending payment	Payment started, awaiting verification	Confirmed, Expired, Cancelled
Confirmed	Payment verified, session scheduled	Completed, Cancelled, No-show (a reschedule keeps it Confirmed)

Status	Meaning	Allowed next states
Completed	Session took place; review and earnings open	Disputed
Cancelled	Cancelled by learner, tutor or admin; refund rules apply	none
Expired	Hold or payment window lapsed; slot released	none
No-show	Learner or tutor absent; policy outcome applied	Disputed
Disputed	Under support review	Completed, Cancelled

Payment states

Initiated → Verified, Failed or Abandoned. Verified → Refund requested → Refund approved → Refund processing → Refunded (full or partial). Verified → Chargeback → Resolved. Each transition writes an immutable ledger event.

Workflow A: Student asks a Mathematics question

1. Student signs in and opens Trainbow HUB AI.
2. Server checks permissions and remaining quota.
3. Gateway sends the question with only that learner's allowed context, applying moderation.
4. Mathematics answers are verified where possible; unverified answers are labelled as such.
5. Response, usage and cost are stored; the student can rate it or ask for a tutor.
6. If the provider fails, the student sees a retry option and no quota is used.

Workflow B: Student books a tutor

1. Student filters tutors and opens an approved profile.
2. Student picks a slot; the server holds it with a database-level lock.
3. Checkout shows price, commission-inclusive total and cancellation terms.
4. Provider payment starts with an idempotency key.
5. Signed webhook and API verification confirm payment.

6. Booking becomes Confirmed; both parties are notified and a classroom link is issued.
7. Reminders go out; attendance is recorded; booking completes by rule (tutor marks done and no learner dispute within the agreed window).
8. Review and earnings entries unlock.

Workflow C: Parent enrols a child

1. Parent registers and verifies phone or email.
2. Parent creates or links the child profile and records consent.
3. Parent chooses a tutor or course and pays.
4. Child receives permitted access only; the parent sees upcoming sessions and progress.

Workflow D: Refund or dispute

1. Authorised user submits a request with a reason.
2. Support reviews the booking, attendance and policy.
3. An authorised approver approves or declines; a second approver is required above a set amount.
4. The provider refund is initiated and its result reconciled.
5. Booking, payment and ledger update together; the customer is notified; the case is kept for audit.

Edge cases to test

- Webhook arrives before the user returns to the site, after it, twice, or never (reconciliation job catches it).
- Tutor cancels after payment: automatic full refund, tutor penalty rules apply.
- Tutor suspended with future bookings: learners are offered rebooking or a refund.
- Learner and tutor are in different time zones or daylight rules change.
- AI conversation deleted while its feedback is under review.

8. Security, privacy, safeguarding and AI governance

These controls must be in place and tested before any real learner uses the platform. Legal items need review by a qualified Nigerian data-protection and child-safeguarding adviser; this PRD does not substitute for that advice.

Security baseline

- Encrypt data in transit and at rest; keep API keys, payment secrets and signing keys in a managed secret store, never in client code.
- Check authorisation on the server for every protected read and write, including object-level checks (can this user see this booking?).
- Defend against injection, XSS, CSRF, broken access control and insecure file upload; scan dependencies in CI.
- Rate-limit login, OTP, booking, payment and AI endpoints.
- Require MFA for admin and support; log all privileged actions.
- Scrub passwords, payment data and personal data from logs and error reports.
- Run an independent security test before launch and after major changes.

Privacy and data protection

- Name a data protection lead and register or assess obligations under the Nigeria Data Protection Act 2023 and Nigeria Data Protection Commission guidance, including any data protection impact assessment and breach-notification duties.
- Keep a data inventory with lawful basis, purpose, owner and retention period for each data type.
- Document consent for minors, including who may give it and how it is withdrawn.
- Seek legal advice on cross-border transfers before using offshore AI, hosting or video services with learner data, and again before international launch.
- Provide access, correction, export and deletion handling within defined time limits.

Data type	Proposed retention (to confirm with counsel)
AI conversations	User-controlled; default 90 days, deletable any time
Quiz attempts and progress	Life of account plus a stated period
Booking and attendance records	Per legal and dispute needs
Financial records	Per Nigerian tax and accounting requirements
Tutor verification documents	Until approval decision plus a short review period
Session recordings	Not recorded by default

Child safeguarding

- Tutor vetting before approval: identity check, credential check, references and a criminal-record or equivalent screen where lawful and available; a written code of conduct signed by each tutor.
- No private chat for minors; all contact runs through platform channels that support review.
- Guardians can join or observe sessions for younger learners.
- A visible report button in the app and classroom; high-severity reports triaged within 24 hours with a named safeguarding lead and written escalation path (including to authorities where required).
- Tutors can be suspended immediately pending investigation, with automatic notice to affected learners.

AI governance

- **Disclosure:** all AI output is labelled and users are told it can be wrong.
- **Grounding and verification:** use approved curriculum content, and verify arithmetic and generated answer keys programmatically.
- **Safety:** moderate inputs and outputs, block off-topic or unsafe content, and keep responses age-appropriate.
- **Privacy:** send the provider the minimum data needed, no names or contact details, and confirm the provider's data-use terms before launch.
- **Quality loop:** review a weekly sample of conversations (with privacy controls), track error reports and update prompts and content.
- **Cost control:** per-user daily limits, global budget cap, automatic degrade or pause at threshold.

9. Quality, delivery and handover

Testing

Type	Must cover
Unit	Pricing, commission, permissions, validation, state transitions

Type	Must cover
Integration	Database, AI, payment, email/SMS and video providers
End to end	Register, enrol, book, pay, attend, review, refund
Concurrency	Many simultaneous attempts on one slot; one winner
Payment	Delayed, duplicate and forged webhooks; failures; partial refunds; reconciliation
Security	Object-level access, authentication, rate limits, OWASP-class issues
AI evaluation	Accuracy against rubric, hallucination, unsafe content, age fit, cost per session
Accessibility	Keyboard, screen reader, contrast, form errors
Performance	Mobile speed on throttled networks, load at expected pilot peak plus margin
Recovery	Backup restore, provider outage behaviour, rollback

Release gates

The MVP does not open to the public until all eight are met:

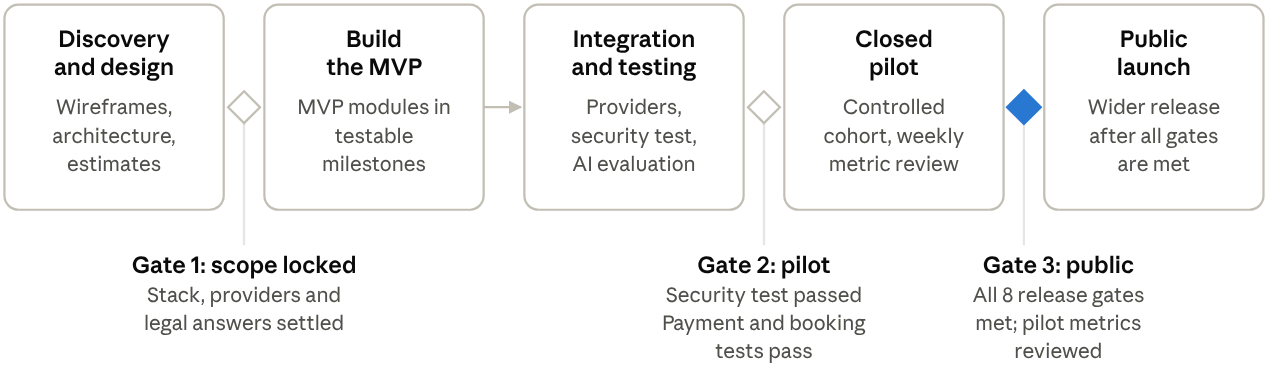
1. No open critical or high security issue.
2. Booking conflicts prevented under concurrency tests.
3. Payment verification and reconciliation pass all scenarios.
4. Student, guardian, tutor and admin access controls tested.
5. AI accuracy, safety and fallback evaluated and signed off.
6. Backup restoration demonstrated in staging.
7. Support, safeguarding and incident procedures documented and rehearsed.
8. Terms, privacy, refund and safeguarding policies published and legally reviewed.

DevOps

Use separate development, staging and production environments with separate credentials. The CI pipeline runs lint, tests and build checks; database migrations are reviewed and reversible. Configure alerts for payment failures, booking errors, AI cost spikes and downtime, plus a rollback procedure and a status-communication plan.

Roadmap

Three gates separate discovery, pilot and public launch



Phases are in order, not to scale. Durations are set after discovery.

roadmap · 5 phases, 3 gates

The 90-day figure from v1.0 is a target for a tightly scoped MVP with an experienced team; the real schedule is set after discovery.

Open questions and decisions

#	Question	Owner	Needed by
1	Which payment provider, and does it support refunds, webhooks and the intended commission model?	Product owner + tech lead	End of discovery
2	What is the legal age threshold and consent method for minors?	Legal adviser	Before design freeze
3	What commission rate and cancellation and refund policy apply by default?	Business owner	Before checkout design

#	Question	Owner	Needed by
4	Which AI provider, and can learner data leave Nigeria under our compliance position?	Tech lead + legal adviser	End of discovery
5	Native learning module or Moodle?	Tech lead	End of discovery
6	Which tutor vetting checks are legally available and practical?	Safeguarding lead	Before tutor onboarding
7	Pilot size, tutors available and target launch date	Business owner	Start of discovery
8	Budget ceiling for monthly AI, video and SMS costs	Business owner	End of discovery

Definition of done

A feature is done only when its acceptance criteria pass; server-side permissions, validation and error states are implemented; loading, empty, success and failure screens exist; tests pass in CI and staging; accessibility and security are reviewed; logs and alerts are configured; help text and documentation are updated; and a rollback path exists for high-risk changes.

Handover

1. Source code in a repository owned by Trainbow.
2. Deployment and environment documentation.
3. Database schema and migrations.
4. API documentation and integration details.
5. Automated tests and reports.
6. Admin, support and tutor guides.
7. Backup and recovery runbook.
8. Security and privacy documentation.
9. Inventory of third-party services and recurring costs.
10. Domain, hosting, payment and AI accounts registered to the business.
11. Known limitations and a prioritised backlog.
12. Staff training for platform operations.

Contracts must state ownership of code, designs, content, credentials and licences.

Vendor brief

Build Trainbow HUB, a secure, mobile-first Nigerian education platform: public catalogue, role-based accounts with guardian consent, Mathematics learning and quizzes, a server-side AI Mathematics assistant with quotas and tutor escalation, a vetted tutor marketplace with concurrency-safe booking, verified NGN payments, refunds and reconciliation, a third-party video classroom, and admin tools. Propose the simplest maintainable architecture, using managed services where sensible.

Proposals must include:

1. Stack and architecture.
2. MVP scope and exclusions.
3. Wireframes or clickable prototype.
4. Milestone timeline and budget.
5. Itemised recurring costs.
6. Security, privacy and safeguarding plan.
7. Test and acceptance plan.
8. Maintenance and support terms.
9. IP and account-ownership terms.
10. Handover and post-launch support.

Pilot-ready means a student can register, learn, use the AI, find a tutor, book, pay, attend and see progress, while parents, tutors and administrators can do their own tasks without exposing anyone else's data or money records.